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BLOCK ROTATIONS IN THE RIO GRANDE RIFT, NEW MEXICO

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Abstract. Additional paleomagnetic data from the north-central Rio Grande rift support previous suggestions for counterclockwise rotation of a large crustal block. Thirteen sites from intrusive and extrusive rocks associated with the Oligocene Espinaso Formation in the area around the Ortiz Mountains, central New Mexico, yield magnetic directions of I = 42.1° and D = 337.8° (alpha-95 = 10.3°). Comparison with the expected direction for the Oligocene indicates 17.8° ± 11.4° of counterclockwise rotation. This agrees with previous studies of Miocene to Pliocene rocks north of the study area that also show about 15° of counterclockwise rotation and corroborates the timing of this motion as post 5 Ma. The extent of the rotated block now includes areas in and adjacent to the Espanola basin east of the Pajarito fault zone, from the San Luis basin in the north to the Albuquerque basin in the south. Extensive published geologic and structural information indicates that counterclockwise block rotation in the Rio Grande rift probably results from left slip along the rift. The rotated block is diamond shaped and bounded by major fault zones (Embudo fault to the north, Pajarito fault zone to the west, Tijeras-Canoncito fault zone to the southeast, and Picuris-Pecos fault to the east). Structural consequences of the counterclockwise block rotation are uplift at the acute ends (Sandia and Picuris uplifts) and subsidence at the obtuse ends (Velarde graben and Santa Fe embayment).

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INTRODUCTION

While it is generally agreed that rifts form in extensional tectonic environments, it is also well known that a number of rifts have experienced significant strike-slip displacements (e.g., Rhinegraben [Illies, 1979]; [Illies and Greiner, 1978]; Gulf of California [Larson et al., 1968]; Rio Grand rift [Kelley, 1979, 1982]; [Muehlberger, 1970]; [Dungan et al., 1984]). Through a series of analytic techniques and clay model experiments the types of faults predicted in a rift environment have been shown to vary systematically with the angle between the extension direction and the rift axis [e.g., Withjack and Jamison, 1984, 1985]. For extension at a significant angle to the rift axis, strike-slip faults and obliquely striking normal faults are expected. Thus the formation of strike-slip faults in rifts should be a fairly common occurrence, given that

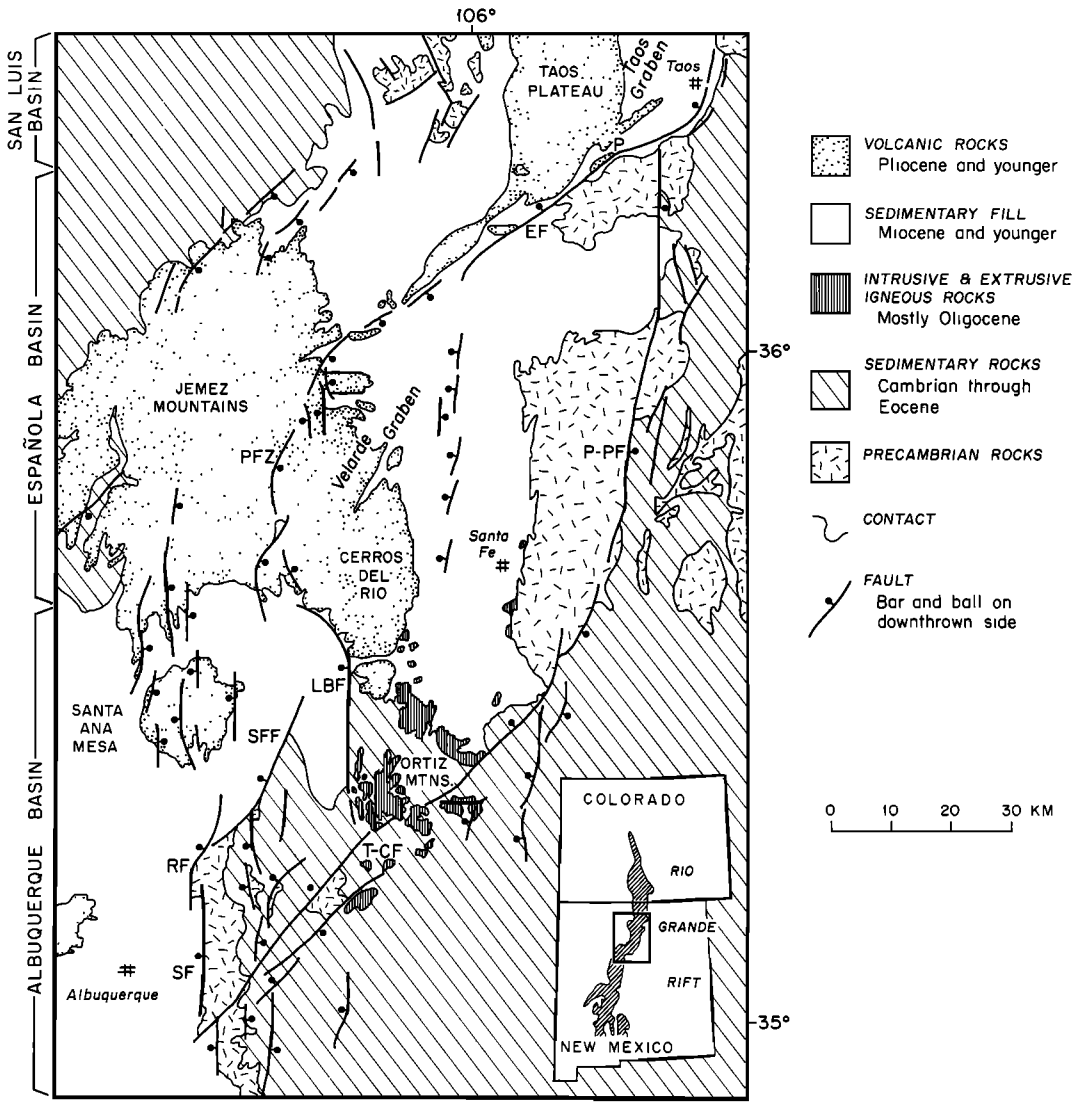


Fig. 1. Generalized geologic map of a portion of the Rio Grande rift in north-central New Mexico. The map includes parts of the San Luis, Espanola, and Albuquerque basins, each of which contains a well-known New Mexico city: Taos, Santa Fe, and Albuquerque. The presently active subbasins of the three basins are the Taos, Velarde, and San Felipe (cutting Santa Ana Mesa) grabens, respectively. Important faults are, from north to south: EF, Embudo fault; P-PF, Picuris-Pecos fault; PFZ, Pajarito fault zone; LBF, La Bajada fault; T-CF, Tijeras-Canoncito fault; SFF, San Francisco fault; RF, Rincon fault; and SF, Sandia fault. Places mentioned in the text are: P, Pilar; Dixon, located near the "F" in EF; Sandia Mountains, Precambrian-cored mountains east of Albuquerque; Sangre de Cristo Mountains, Precambrian-cored mountains extending north from just south of Santa Fe into Colorado; Hagan embayment, a low between SFF and LBF; Santa Fe embayment, a low between the Ortiz Mountains and the southern Sangre de Cristo Mountains.

the extension direction will not always lie perpendicular to the rift axis. Recent studies also show that a significant amount of strike-slip motion distributed amongst a number of faults will result in the rotation of tectonic blocks between faults [e.g., Ron et al., 1984; Garfunkel and Ron, 1985]. As a result, the rotation of crustal blocks should be a fairly common occurrence

within rifts that are undergoing some component of strike-slip motion.

In a previous paper [Brown and Golombek, 1985] we show the existence of significant clockwise and counterclockwise block rotations in the Espanola basin of the Rio Grande rift. Paleomagnetic studies on late Tertiary rocks on either side of the Pajarito fault zone (Figure 1) show magnetic directions 10°-20° from the expected direction. Specifically, rocks west of the Pajarito fault zone are rotated 15° clockwise, whereas rocks east of the fault zone are rotated counterclockwise a similar amount. These results are based on paleomagnetic studies of four volcanic units in the Jemez Mountains (Paliza Canyon Formation, Lobato Basalt, Tschicoma Formation [Brown and Golombek, 1985] and Valles Rhyolite [Doell et al., 1968]) and three studies east of the fault (Chamita Formation [MacFadden, 1977], Tesuque Fromation [Barghoorn, 1981], and Cerros del Rio Volcanics [Brown and Golombek, 1985]). Because the rotations of Miocene and older rocks are roughly the same, the initiation of rotation must have begun after the Miocene, about 5 Ma. Differential rotation across the fault zone is about 4° per million years. The rotations are interpreted to be due to (1) clockwise rotation of the Colorado Plateau and/or rotation of a small block or blocks in a WNW-ESE extensional stress field [e.g., Golombek et al., 1983] and (2) counterclockwise rotation within the Rio Grande rift due to significant left slip.

In this paper we report on paleomagnetic data that help define the areal extent and timing of a counterclockwise-rotated tectonic block in the Rio Grande rift [Golombek and Brown, 1983; Brown and Golombek, 1984]; a rift along which significant strike-slip motion has previously been suggested to have occurred.

TECTONIC SETTING

The intraplate Rio Grande rift is composed of a number of extensional basins that form a valley system from central Colorado to southern New Mexico [e.g., Chapin, 1979]. The rift is an important element in the tectonic origin and evolution of the western United States that includes four major physiographic provinces: the Basin and Range, Colorado Plateau, Southern Rocky Mountains, and Great Plains.

In northern New Mexico, three basins make up the Rio Grande rift [Kelley, 1979]: The Albuquerque, Espanola, and San Luis basins (Figure 1). These basins are north-trending, asymmetric, fault-bounded basins that are filled with predominantly syn-rift Miocene sedimentary fill (Santa Fe Group). Santa Fe Group sediments in these basins are capped by volcanic fields, from south to north in Figure 1: Albuquerque, San Felipe (capping Santa Ana Mesa), Cerros del Rio, Jemez, and Taos Plateau volcanic fields. The Ortiz Mountains are made up of intrusive and extrusive igneous rocks that were emplaced before and during the initial stages of Rio Grande rift development [Baldridge et al., 1980].

Most recent tectonic activity in the Taos, Espanola, and northern Albuquerque basins has been along central grabens [Muehlberger, 1979; Golombek, 1981; Dungan et al., 1984; Kelley, 1977] that are from north to south (Figure 1): the Taos, Velarde, and San Felipe (cutting Santa Ana Mesa) grabens. Like the basins, these grabens are north-trending depressions that step en echelon to the right. Between the Velarde and Taos grabens the step occurs by pivoting across the recently active Pajarito fault zone and Embudo fault [Kelley, 1979; Muehlberger, 1979].

GEOLOGIC SETTING

Intrusive and extrusive igneous rocks of mid-Tertiary age crop out along the eastern edge of the Rio Grande rift between the Sandia Mountains to the south and the Sangre de Cristo Mountains to the north (Figure 1). These outcrops include intrusive rocks in the Ortiz Mountains and Cerrillos Hills and extrusive flows in the Hagan embayment, in the Cienega area, and in Arroyo Hondo south of Santa Fe. The intrusive rocks are found within a sequence of prerift Mesozoic and early Cenozoic sedimentary rocks. Extrusive rocks overlie the Eocene Galisteo Formation [Stearns, 1943], and in turn are overlain by rift sediments of the Santa Fe Group. Igneous activity appears to have occurred prior to rifting, although the youngest volcanics may be contemporaneous with the early stages of rifting.

Stearns [1943] named the collective extrusive rocks the Espinaso Volcanics.

TABLE 1. Published Ages, Espinaso Formation and Associated Igneous Units

Material and Location	Age, Ma	Reference*
Glassy latite, Espinaso Formation, La Cienega	19.5 ± 0.5 19.6 ± 0.6	1
Andesite flow, Arroyo Hondo, Santa Fe	24.9 ± 0.6 25.3 ± 0.6	2
Cieneguilla Limburgite, La Cienega	25.1 ± 0.7	1
Olivine-tholeiite flow, near base of overlying Santa Fe Formation	25.1 ± 0.6	3
Clast, Breccia unit, Espinaso Formation La Cienega	25.8 ± 0.9	1
Latite flow, near top of Espinaso Formation	26.9 ± 0.6	3
Andesite, Espinaso Formation, Arroyo Hondo	29.3 ± 0.6	1
Augite monzonite porphyry, La Cienega	30.2 ± 0.7	1
Latite sill, west side Ortiz Mountains	34.0 ± 2.2	2
Latite clast, near base of Espinaso Formation	34.3 ± 0.8	3
Clast, near middle of Espinaso Formation	34.6 ± 0.7	3
Latite porphyry cobble, Espinaso Formation	36.9 ± 1.2	4
Latite intrusion, southern Cerrillos Hills	47.1 ± 3.2	2

*Sources of dates: 1, Baldridge et al. [1980]; 2, Bachman and Mehnert [1978]; 3, Kautz et al. [1981]; 4, Weber [1971].

In later publications, Stearns [1953a, 1953b] described the extrusive rocks and related them to the nearby intrusions. He also named the Cieneguilla Limburgite, a series of basalt flows exposed north of Cienega. Associated with the Espinaso Volcanics are water-laid breccias, tuffs, and conglomerates of predominantly volcanic material. Later work indicated that much of the unit is actually sedimentary and the name was changed to Espinaso Formation [Bachman, 1975; Baltz, 1978]. Kautz et al. [1981] present a complete sedimentary and petrologic study of the formation.

The intrusive rocks associated with the Espinaso Formation have not been studied in great detail, although several areas have been mapped and investigated since Stearns' early work. Sun and Baldwin [1958] described the Cienega area, Disbrow and Stoll [1957] described the Cerrillos Hills, and Bachman [1975] mapped the Madrid 15' quadrangle, which includes the Ortiz Mountains. Intrusive plugs, sills, and stocks of predominantly monzonite porphyry are associated with numerous dikes of intermediate composition. Disbrow and Stoll [1957] recognized four periods of igneous activity in the Cerrillos Hills, based on field evidence and composition. However, the intrusive rocks of the Ortiz Mountains and Cienega area are all mapped together as one unit.

Radiometric age dates for intrusive, extrusive, and detritus related to the Espinaso Formation indicate a mid-Tertiary age. Fifteen K-Ar dates, with rock type, locations, and references are listed in Table 1. With the exception of the youngest (19.5 Ma) and oldest (47.1 Ma) dates, the ages are all Oligocene (38 to 25 Ma). Despite the numerous dates available, it is not possible to establish a detailed chronology for the various Espinaso intrusives and extrusives. Age data on the Cieneguilla Limburgite [Baldridge et al., 1980] indicate that these flows are probably contemporaneous with latest Espinaso activity, although Stearns [1953b] and Sun and Baldwin [1958] report an angular unconformity below the limburgite.

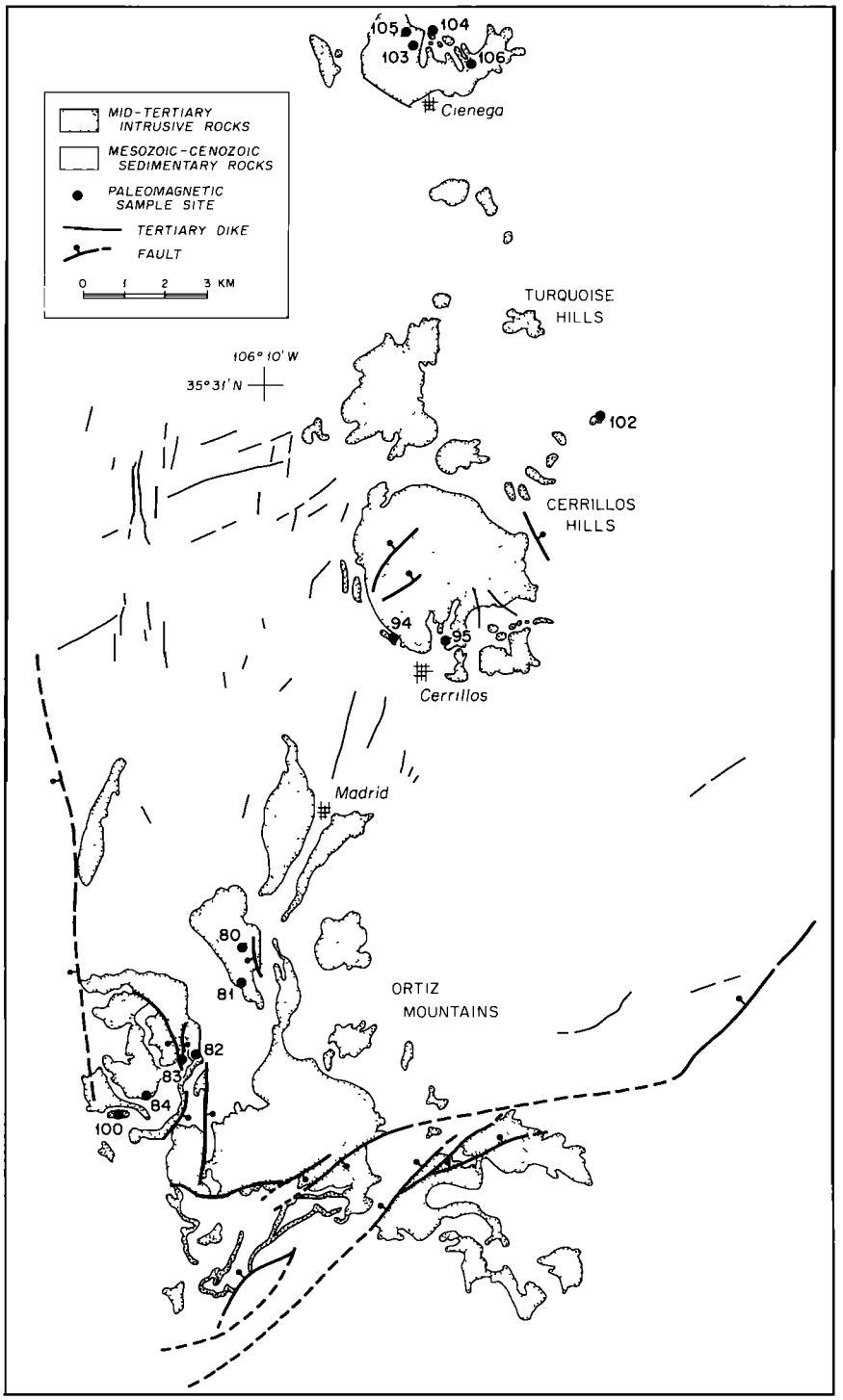


Fig. 2. Simplified geologic map of the Ortiz Mountains and surrounding area from Bachman [1975], Disbrow and Stoll [1957], and Sun and Baldwin [1958]. Paleomagnetic sample sites are indicated by solid black circles. Note that two sites (97 and 99) are in the Hagan embayment off the diagram to the southwest.

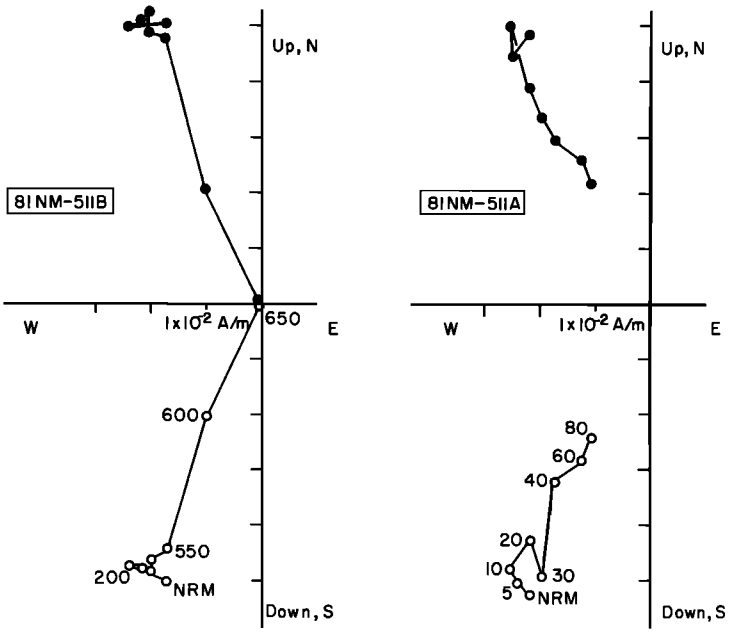


Fig. 3. Vector diagrams of demagnetization for two specimens from one sample, site 81. Open circles are projections on the vertical plane; closed circles are on the horizontal plane. (a) Alternating field demagnetization, steps in milliteslas. (b) Thermal demagnetization, steps in degrees centigrade.

PALEOMAGNETIC DATA

Field Sampling

Fifteen sites were collected from intrusive and extrusive rocks associated with the Espinaso Formation. Ten sites were in the intrusive units; six sites were located south of Madrid, on the western side of the Ortiz Mountains, (sites 80-84, 100), three sites were in the Cerrillos Hills north of Cerrillos (sites 94, 95, 102), and one site was northwest of Cienega (site 103) (Figure 2). The Cieneguilla Limburgite was sampled at three sites north of Cienega (sites 104-106). Also sampled were two sills in the Hagan embayment (sites 97, 99), southwest of the area depicted in Figure 2.

Sites were drilled in the field, using a gasoline-powered rock drill, and oriented with magnetic and sun compasses. In most cases, sun and magnetic compass readings agreed within 2°, although, in a few cases, larger variations were recorded.

The massive nature of the intrusive rocks in the Cerrillos Hills and Ortiz Mountains precluded observations of local strikes and dips at sampling sites. It is evident from geologic mapping in the region that the Oligocene rocks were intruded into a preexisting synclinal structure [Stearns, 1953b]. There is no convincing evidence in the immediate area indicating a regional tilt since the time of intrusion. There has been recent faulting in the Hagen basin, and the sills in the Hagen area (sites 97 and 99) were corrected using strikes and dips collected at each site. Sun and Baldwin [1958] observed a shallow (10°) northeast dip in the limburgite, and this was used to correct sites 105 and 106.

Numerous dikes were also investigated in the course of the field work. Many of the dike outcrops were severely weathered, making drilling impossible and sampling undesirable. Several dikes were drilled in the field, but samples proved to be magnetically unstable during measurement and were abandoned as accurate paleomagnetic recorders.

Laboratory Procedures and Results

Natural remanent magnetization (NRM) directions were obtained for all samples, using a Schonstedt spinner magnetometer coupled to a NorthStar computer. NRM directions are quite scattered but

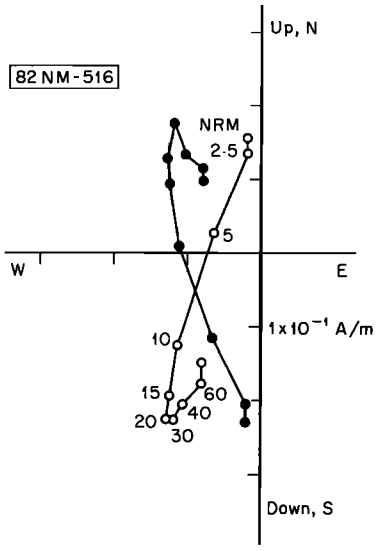


Fig. 4. Vector diagram of alternating field demagnetization for sample from site 82. Symbols as in Figure 3.

indicate a preference for north to north-northwest declinations and intermediate positive inclinations. A small group of reversed directions (negative inclinations, south to southeast declinations) is also present. Additionally, there is a cluster of directions with east declinations and shallow to intermediate positive inclinations.

Step demagnetization studies were done on selected samples from each site, using both thermal and alternating field techniques. Samples from the intrusive rocks show two types of demagnetization behavior. Cores with initial directions close to the expected Oligocene direction tend to have demagnetization paths indicating one component of demagnetization and straight decay curves (Figure 3). This type of behavior is seen in both thermal and alternating field studies on specimens from the same core. NRM intensities on these samples ranged from 5×10^{-1} to 5×10^{-3} A/m. Median destructive fields were in the range 40-60 mT, although many of the samples had appreciable magnetization remaining after demagnetization in fields of 80 to 100 mT. Thermal studies on the same rocks reveal similar behavior with little direction change and little loss of intensity until temperatures greater than 550°C. Reversed polarity samples in this group also behaved in a similar way, maintaining stable consistent directions during demagnetization. In a few instances, reversed NRM directions reverted to normal directions after demagnetization (Figure 4). The remaining samples for sites in this group were demagnetized in alternating fields of 40 to 60 mT.

A second set of intrusive samples had initial directions with easterly declinations. In pilot samples, mean coercivities were low (5 to 10 mT) and directions changed rapidly to north-northwest declinations, with inclinations remaining intermediate and positive (Figure 5). Thermal studies on the same cores also produced direction changes but not as fast or as complete as the alternating field treatments. Samples in this group (sites 80, 102, and part of 81) all had higher initial NRM intensities (40 to 1 A/m) than the other intrusive samples. The strong easterly component was removed by demagnetization fields of 20 to 40 mT, and the remaining samples were demagnetized at 40 mT. Site 103, from the intrusive rock northwest of Cienega, shows scattered easterly NRM directions. Even after demagnetization these directions remained unchanged and scattered (alpha-95 = 34.0° at 40 mT).

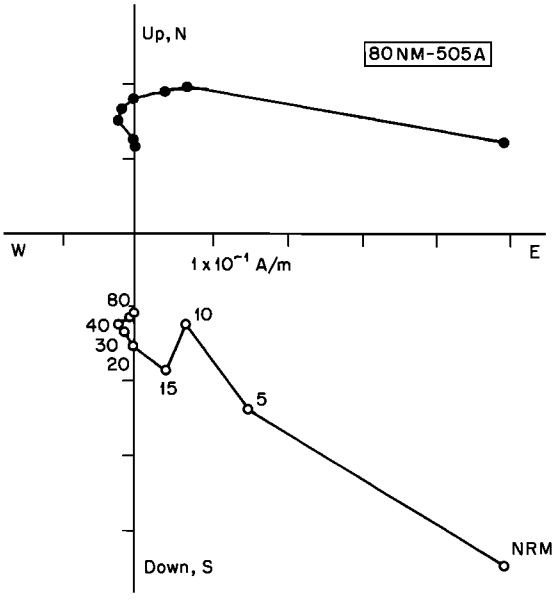


Fig. 5. Vector diagram of alternating field demagnetization for sample from site 80. Symbols as in Figure 3.

TABLE 2. Paleomagnetic Directions, Oligocene Igneous Units, Central New Mexico

Site	N	I	D	K	a-95	POLE Latitude	POLE Longitude
80	6/6	42.1	2.6	86	7.3	78.7	61.6
81	7/7	43.0	341.5	34	10.5	71.0	135.8
82	7/7	50.2	323.2	120	5.5	59.1	166.3
83	4/7	48.8	320.0	16	24.1	56.0	165.8
84	5/5	51.1	347.9	107	7.4	79.3	147.7
94	6/6	-50.7	106.3	36	11.3	29.8	183.1
95	5/6	-14.1	173.7	90	8.1	61.1	86.9
97	5/8	-27.0	159.8	12	23.6	62.2	119.6
99	4/6	28.5	351.9	14	18.7	68.6	95.6
100	7/7	-44.2	160.8	109	5.8	71.0	139.3
102	5/7	29.2	309.0	17	19.2	40.5	153.8
*103	6/6	32.0	111.7	6	34.0	6.6	137.1
*104	5/6	1.0	24.0	12	23.4	48.4	36.1
105	4/6	41.8	0.2	101	9.2	78.5	72.9
106	6/6	51.6	336.0	210	4.6	72.7	152.7
Mean	13	42.1	337.8	17	10.3		
				15	11.1	67.9	142.3

N, number of samples used/number of samples collected; I, inclination, in degrees; D, declination, in degrees; K, precision parameter; a-95, radius of cone of 95% confidence about direction, in degrees; Pole, virtual geomagnetic pole; latitude in degrees north; longitude in degrees east. Sites 97, 99, 105, and 106 are field corrected for local tilting.

*Excluded from mean calculation.

Mean characteristic directions from the igneous units in the Ortiz Mountains area are given in Table 2, with an overall mean of I = 42.1° and D = 337.8° (N = 13, alpha-95 = 10.3°). Included in Table 2, but excluded from the overall mean calculation, are sites 103 and 104; both sites appear to record scattered transitional directions. The 13 sites used in the resulting mean, with its corresponding circle of 95% confidence, are plotted in Figure 6.

ROTATIONS

Rotations (horizontal movements about a vertical axis) and flattenings (vertical movements about a horizontal axis) are determined by comparing paleomagnetic data from one area to the expected direction for that region [Beck, 1980]. Table 3 shows rotation and flattening data for Ortiz Mountains area rocks with respect to three Oligocene poles for North America. Pole A is the 30-Ma pole of Harrison and Lindh [1982] determined from 29 studies using a 20 m.y. sliding window. Pole B is calculated from data given by Irving and Irving [1983] for Oligocene poles for North America. Five studies were used (21-25), representing 24 to 36 Ma. Pole C is from Diehl et al. [1983], determined using Oligocene to early Miocene studies and weighting each site individually. As can be seen in Table 3, expected directions at the latitude of the study area are not markedly different for the three poles. We will use Pole B in further discussion, as it is the most commonly cited source for North American pole positions.

Data in this study show 17.8° ± 11.4° of counterclockwise (negative) rotation with respect to Pole B (Table 3). This movement is similar in amount and direction to other units in the eastern Espanola basin, as reported by Brown and Golombek [1985]. The overlying Santa Fe Group shows -11.5° ± 5.3° for the Chamita Formation and -18.5° ± 5.8° for the Tesuque Formation. The younger Cerros del Rio volcanic field shows -17.1° ± 20.4°, not statistically distinct at the 95% level but the same direction and magnitude. The units east of the Pajarito

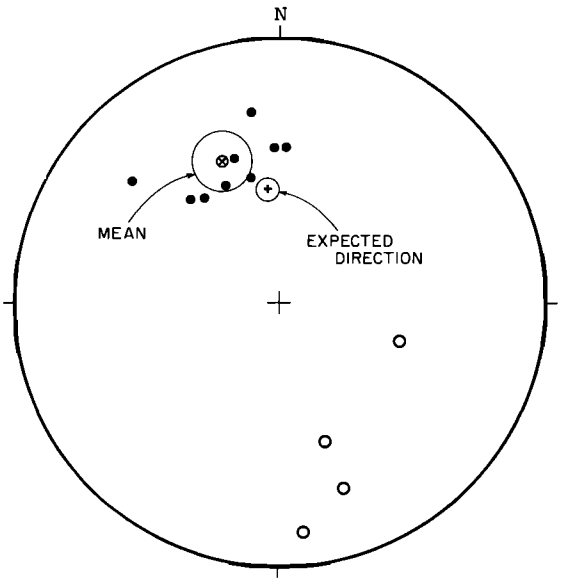


Fig. 6. Equal-area plot of site means after demagnetization. Closed circles are on lower hemisphere; open circles on upper hemisphere. Data listed in Table 2. Mean for Ortiz Mountains area igneous rocks shown by circled cross. Circled plus is expected direction for Oligocene pole of Irving and Irving [1983]. Both mean directions are shown with circles of 95% confidence; see Table 3 for data.

fault zone, treated as a group, yielded a counterclockwise rotation of -16° ± 10°. The rocks in the Ortiz Mountains area south of Cerros del Rio have been rotated the same amount and are probably part of the same block as the units mentioned above (see later discussion).

In our previous paper [Brown and Golombek, 1985] we suggested that the rotation was fairly recent, occurring in the past 5 m.y., based on the similarity in rotations between 5- to 10-Ma rocks and smaller rotations in 2.5- and 1-Ma rocks. The data presented herein agree with this earlier interpretation. The Oligocene rocks show no more rotation than the Miocene-Pliocene units and so must have been rotated after all the rocks were deposited, or since the Pliocene.

It is important to note that these data also show an approximate flattening of 10°. Although it is not statistically distinct (calculated errors are equal to or greater than the flattening), it does suggest the possibility that the Ortiz Mountains area data is 10° too shallow and perhaps has been tilted to the south. Southward tilting of this area would be expected from the predominantly dip slip motion (down to the north) during the later Cenozoic on the Tijeras-Canoncito fault system [Lisenbee et al., 1979], which lies just to the south of the sampled area.

DISCUSSION: LEFT SLIP AND STRUCTURE

Paleomagnetic data that indicate counterclockwise rotations of crustal blocks fit into tectonic models for the Rio Grande rift that include some component of left slip. Tectonic models suggesting such motions, in fact, have

TABLE 3. Rotation Statistics, Oligocene Igneous Units, Central New Mexico

Study	N	I	D	a-95	POLE Latitude	POLE Longitude	A-95	R	ΔR	F	ΔF
Ortiz	13	42.1	337.8	10.3	67.9	142.3	11.1				
Pole A	29	54.2	353.6	...	84.7	157.7	3.6	-15.8	10.4	12.1	12.1
Pole B	5	52.0	355.6	...	85.4	127.1	6.2	-17.8	11.4	9.9	12.9
Pole C	102	52.8	352.2	...	83.2	148.0	4.1	-14.4	10.5	10.7	12.3

Column heads as in Table 2, with A-95 radius of cone of 95% confidence about the mean pole, in degrees; R, rotation with respect to Pole A, B, or C; and F, flattening with respect to Pole A, B, or C, all in degrees, from Beck [1980]; ΔR and ΔF, errors on rotation or flattening in degrees, corrected using method of Demarest [1983]. Pole A is 30 m.y. pole for North America from Harrison and Lindh [1982]. Pole B is from Irving and Irving [1982], see text for discussion. Pole C is from Diehl et al. [1983], calculated using site means. I and D from poles are calculated for location of Ortiz Mountains.

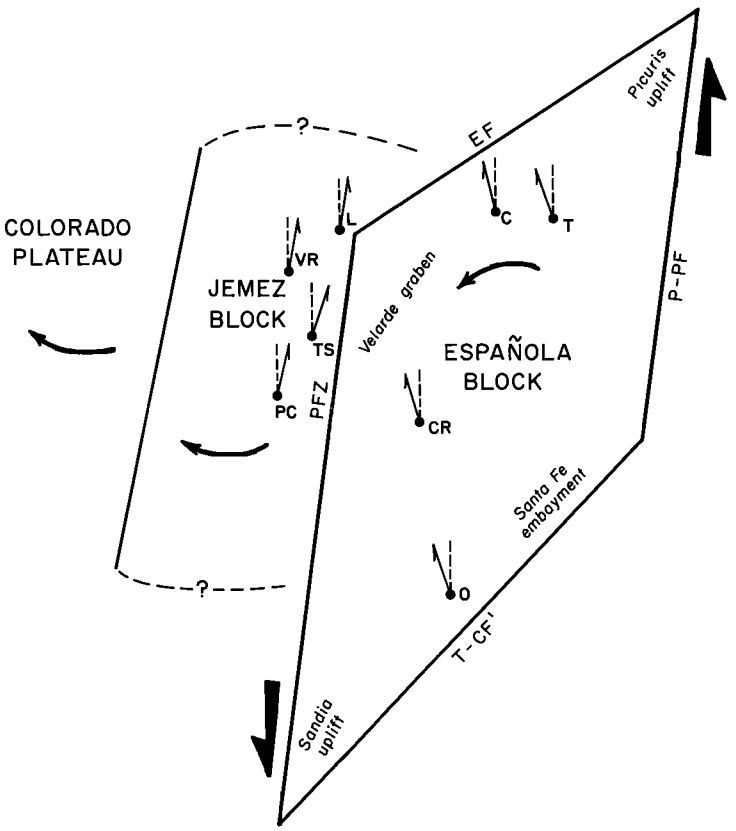


Fig. 7. Schematic diagram of block rotations in the Rio Grande rift. Previously suggested left slip (large arrows) along the Rio Grande rift [Kelley, 1977, 1979, 1982], has caused counterclockwise rotation of the large diamond-shaped "Espanola block." Rotation of the block bounded by important fault zones in the region (defined in Figure 1) has resulted in uplifted highs (Picuris and Sandia uplifts) at the actue ends and downdropped lows (Velarde graben and Santa Fe embayment) at the obtuse ends, similar to the suggestion by Muehlberger [1979]. The "Jemez block" is rotating in a clockwise sense, similar to the expected rotation of the Colorado Plateau [Brown and Golombek, 1985]. Small arrows indicate calculated rotations plotted with expected declinations as north (dashed lines). Studies are: C, Chamita Formation; CR, Cerros del Rio volcanics; L, Lobato Basalt; O, Ortiz Mountains area (this study); PC, Paliza Canyon Formation; T, Tesuque Formation, TS, Tsichoma Formation; VR, Valles Rhyolite. Rotation data for all studies other than the Ortiz Mountains area are from Brown and Golombek [1985].

been proposed previously for this section of the Rio Grande rift, based on a considerable amount of geologic and structural information [Kelley, 1979; 1982; Muehlberger, 1979; Dungan et al., 1984]. This independent geologic information provides strong support for both the timing and size of the rotated block determined by our paleomagnetic work. Although some of this information was briefly discussed in our previous paper [Brown and Golombek, 1985], these paleomagnetic data suggest the size of the rotated block should be enlarged, thereby warranting a more detailed discussion of the structure and tectonics of the block.

Kelley [1952] initially suggested that the en echelon arrangement of the rift basins argued for some lateral component of motion along its length. After considerable study, Kelley [1977, 1979, 1982] documented the right en echelon steps of the basins, their northward ramping and shallowing, and the existence of relay-connecting faults. Along many of these and other border faults of the rift he noted drag and associated flexure

zones and folds that indicate left slip. In addition, Jurassic strata on either side of the rift are offset roughly 30 km, compatible with substantial left slip [Kelley, 1979, 1982].

Muehlberger [1979] and later Dungan et al. [1984] noted from study of the Embudo fault zone that it was acting like an intracontinental transform fault between the actively extending Taos and Velarde grabens of the San Luis and Espanola basins, respectively. The N60E trend of this transform implies a component of left slip along it and the rift. Furthermore, recent compression across the Embudo fault led Muehlberger [1979] to define a diamond-shaped crustal block that was rotating in a counterclockwise sense, thereby developing highs at the acute ends (Picuris and Sandia uplifts) and lows at the obtuse ends (Velarde graben and Santa Fe embayment) (Figure 7). This block is defined by the following major faults (Figures 1 and 7): The Embudo fault to the north, the Picuris-Pecos fault to the east, the Tijeras-Canoncito fault to the south, and the Pajarito fault zone and other faults to the west. A detailed discussion of these faults follows. Note that because this diamond-shaped block is large (roughly, 50 km wide by 150 km long), only a few degrees of rotation could occur without large gaps opening up near the edges of the block or significant internal deformation resulting [e.g., Brown and Golombek, 1985]. Our paleomagnetic data argue for fairly large (~15°) counterclockwise rotations, suggesting that smaller regions within the block could be taking up some of the rotation. Nevertheless, the fact that the deformation expected around the edge is, in fact, observed (see following discussion) implies that at least some of the rotation is accommodated by rotation of the large block just defined.

Embudo Fault

The Embudo fault is a N60E-trending zone that separates the uplifted Precambrian rocks of the Picuris Mountains from the Neogene and younger sediments filling the San Luis basin (Figure 1). This fault zone began with normal displacements that were down to the northwest and later switched to more complex displacements involving both normal and reverse motions [Dungan et al., 1984]. Southwest of Pilar the fault zone appears to be opening and thus under extension. To the northeast of Pilar, however, thrusting and compressional deformation is indicated by microfaults with reverse and strike-slip displacements in Santa Fe Group sediments and northeast-trending anticlines and synclines (including the Gorge arch) in the basalts of the Taos Plateau volcanic field that indicate NW-SE compression in the past 5 m.y. [Muehlberger, 1979]. Stratigraphic and sedimentary-clast provenance analysis shows that compressional deformation is synchronous with uplift of the Picuris Mountains that began in the Pliocene, in agreement with counterclockwise block rotation beginning about 5 Ma and subsequent uplift of this acute end of the block.

In the northeastern Espanola basin, southwest of Pilar near Dixon, Steinpress [1981] suggests a similar history for this area near the Embudo fault. Slickensides, fault drag, and displacements all show left slip motion. Compressional folding and shearing occur along a dextral bend in one fault, as would be expected if the fault was undergoing left slip. In addition, at least one multi-slickensided fault surface shows the most recent deformation to be strike slip with an earlier phase of normal motion. Finally, Steinpress [1981] suggests that a major phase of deformation occurred in the Pliocene from 4.5 to 2.8 Ma, similar in timing to the compressional phase of motion northeast of Pilar.

Pajarito Fault Zone

Detailed geologic and structural work [Golombek, 1981, 1983] as well as our previous paleomagnetic work [Brown and Golombek, 1985] show that the Pajarito fault zone is a complex structural element separating areas with opposing rotations. To a first order, the post-Miocene age of the Pajarito fault zone and the Velarde graben agree well with the suggested 5 m.y. initation age for block rotation and thus support the recent development of a low at this obtuse corner of the block. However, rocks located to the west of the Pajarito fault zone have been rotated clockwise in addition to the counterclockwise rotation of rocks to the east. Although this was noted in our previous paper, reconciling the rotations indicated by our

paleomagnetic data with the detailed mapping has not been attempted and is discussed below.

In general, information on the structure and tectonics of the Pajarito fault zone [Golombek, 1983] does not require rotational or strike-slip motions, however, certain features may be explainable with such motions. As discussed by Golombek [1983], there are a number of northwest-trending faults that connect long north-trending faults in the Guaje Mountain segment of the Pajarito fault zone. These northwest-trending faults could result from left slip locally reorienting the extension direction to WSW-ENE. Mesoscopic fracture data from the Bandelier tuff have been used to derive the extension direction in the past 1.1 m.y. since the tuff was deposited [Golombek, 1981, 1983]. Slickensides show extension both parallel and perpendicular to the local trend of the fault zone. The most straightforward interpretation (prior to paleomagnetic information showing rotations) is that the Pajarito fault zone reoriented the regional minimum and intermediate stress directions to perpendicular and parallel, respectively, to local trend of the fault zone and that the intermediate stress direction was tensional. Other explanations are also possible, however. The generally northerly trending extension direction could result from counterclockwise rotation of the interior crustal block under question. Unfortunately, this can only be suggested qualitatively because the direction of extension due to rotation is critically dependent on both the center of rotation of the block and the number of independent units rotating within the block. Furthermore, the effect of clockwise rotation to the west of the Pajarito fault zone cannot be evaluated.

Other Faults South of the Pajarito Fault Zone

South of the Pajarito fault zone the block that has undergone rotation is bounded by a number of fault zones that are arranged primarily en echelon to the right. These faults (from north to south), La Bajada, San Francisco, Rincon, and Sandia faults (Figure 1), are discussed below.

La Bajada Fault. The La Bajada fault is a north to north-northwest trending, down-to-the-west fault that forms the primary escarpment that separates the Albuquerque-Belen basin from the Espanola basin. The Cerros del Rio volcanic field lies on the up-side of the fault [Aubele, 1979], which cuts a basalt flow dated at 2.5 Ma. To the south this fault separates the Santa Fe embayment from the Hagan embayment [Kelley, 1979]. Paralleling the La Bajada fault to the west is the San Francisco fault, which forms the west side of the Hagan embayment.

San Francisco Fault. The San Francisco fault is a north to northeast-trending, down-to-the-west fault that bounds the eastern side of the Albuquerque-Belen basin at this latitude. This fault juxtaposes Pennsylvanian through Tertiary rocks on the east against Neogene Santa Fe Group on the west [Kelley, 1977]. This fault can be connected with the La Bajada fault to the northeast by a group of east-trending mostly down-to-the-north faults. To the south the San Francisco fault merges with the Rincon fault.

Rincon and Sandia Faults. The Rincon and Sandia faults are the main range-bounding faults between the Sandia uplift and the Albuquerque-Belen basin (Figure 1). These north trending, down-to-the-west faults place Precambrian rocks of the Sandias on the up-side against Tertiary Santa Fe Group sediments on the down-side [Kelley, 1977]. The most southern of these faults, the Sandia fault, intersects the Tijeras-Canoncito fault system to the south.

Tijeras-Canoncito Fault System

The Tijeras-Canoncito fault system is a major northeast-trending transverse structure that delineates the eastern side of the Sandia uplift, cuts through the southern Ortiz Mountains, and forms the southern structural boundary of the Santa Fe embayment (Figure 1). This structure is composed of a system of faults up to 10 km in width that are sigmoidal, anastomosing, and en echelon. The faults within the system are probably vertical and have undergone motion from the Precambrian to recent [Lisenbee et al., 1979]. In addition, they may have influenced emplacement of the Tertiary intrusions and associated mineral deposits [Woodward, 1984]. Latest Cenozoic motion has been predominantly dip slip, with a later phase of strike-

slip motion. The strike-slip motion appears to have both left-lateral and right-lateral components [Lisenbee et al., 1979]. Slickensides, offset of a dike, and the en echelon pattern of a group of faults all indicate left slip [Kelley and Northrop, 1975], whereas the en echelon pattern of some folds imply right slip. Lisenbee et al. [1979] suggest that these phases of motion can result from putting a fault zone with a complicated history of development under extension. However, it is also quite possible that the left-lateral displacements result from the left slip along the rift, while right slip results from counterclockwise rotation of the interior crustal block.

Uplift of the Sandias might be expected from counterclockwise rotation of the central block in question, if the area to the east were a stable buttress. It is interesting to note that fission track data do, in fact, show rapid uplift of the Sandias since late Miocene [Kelley and Duncan, 1984], in agreement with the proposed recent block rotation.

Picuris-Pecos Fault

The Picuris-Pecos fault is a major north-trending fault that forms the eastern side of the crustal block, intersecting the Trijeras-Canoncito fault to the south and the Embudo fault to the north (Figures 1 and 7). This fault cuts roughly 100 km through the Sangre de Cristo Mountains and has had recurrent motion since the Precambrian [Montgomery, 1953; Miller et al., 1963]. Approximately 23 km of right-lateral slip has occurred in the Precambrian. Vertical motion is documented in the Paleozoic (east side down) and Laramide (west side up). This fault was also active (east side down) when the Rio Grande rift was opening in the Cenozoic, however, no work has been reported on that attempts to unravel the latest stages of motion.

Summary

Geologic and structural information pertaining to the major bounding faults and the direction, structural consequences, and timing of the rotation all independently support the rotations indicated by our paleomagnetic data. A diamond-shaped block initially proposed by Muehlberger [1979] is defined by the following major fault zones: the Embudo fault on the north, the Pajarito fault zone and La Bajada, San Francisco, Rincon, and Sandia faults on the west, the Tijeras-Canoncito fault system to the south, and the Picuris-Pecos fault to the east. Most of these faults show evidence for activity since the Miocene, roughly when the paleomagnetic data indicate rotation began. Structural consequences of the proposed counterclockwise block rotation are uplift at the acute ends (Picuris and Sandia mountains) and subsidence at the obtuse ends (Velarde graben and Santa Fe embayment). Geologic and geophysical data show uplift of the Picuris Mountains in the Pliocene, rapid uplift of the Sandia Mountains since the late Miocene, and subsidence of the Velarde graben since the early Pliocene, all independently supporting the block rotation. Finally, the cause of the block rotation appears to be significant left slip along the rift, which has also been previously suggested on the basis of geologic and structural information.

CONCLUSIONS

1. Paleomagnetic data from intrusive and extrusive rocks associated with the Oligocene Espinaso Formation around the Ortiz Mountains, New Mexico, yield mean magnetic directions of $I = 42.1°$ and $D = 337.8°$, with a corresponding pole position of 67.9°N latitude and 142.3°E longitude. A comparison with the expected Oligocene direction indicates $17.8° \pm 11.3°$ of counterclockwise rotation.

2. The observed rotation agrees in direction (counterclockwise) and magnitude (~15°) with the rotation determined for Miocene and younger rocks east of the Pajarito fault zone [Brown and Golombek, 1985]. This similar rotation for Oligocene to Pliocene age rocks supports our previous suggestion that motion has occurred in the past 5 m.y.

3. The location of these rotated rocks near the Ortiz Mountains south of previously studied areas enlarges our concept of the block undergoing counterclockwise rotation. These data support the diamond-shaped crustal block proposed by Muehlberger [1979], extending from the San Luis basin southward to the Albuquerque basin. This block is bounded on all sides by well-defined, recently active fault zones.

4. The size of the block, the

counterclockwise nature of rotation, and the recent timing of motion (since 5 Ma) are all independently supported and corroborated by geologic and structural data. Left slip along the rift, indicated by abundant geologic and structural evidence, probably provides the driving mechanism for the counterclockwise rotation. Coutnerclockwise rotation of the block has resulted in the development of topographic highs at the acute ends of the block (Sandia and Picuris uplifts) and lows at the obtuse ends (Velarde graben and Santa Fe embayment).

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